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Implementation of a Free Software for Managing Imaging Protocols

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1. Introduction

According to the document from the American Association of Physicists in Medicine (AAPM, 2013), the enhancement and management of CT protocols are fundamental to ensuring diagnostic quality, optimizing radiation dose, and guaranteeing the proper use of CT scanners, resulting in patient safety. Institutions performing CT exams operate with various tomographic exam protocols, defined according to the anatomical region of diagnostic interest, clinical indication, patient age group, contrast agent to be used, among other particularities. Given the complexity and diversity of aspects to be considered in the exams, the process of protocol optimization requires full knowledge of their specificities. This study aims to describe the characteristics and frequencies of CT exams performed in a university hospital, providing support for improvements in protocol management and optimization strategies.

2. Materials and Methods

A retrospective study was conducted in the Computed Tomography (CT) Department of the Hospital das Clínicas at the Federal University of Pernambuco (HC-UFPE), based on data extraction carried out using the open-source software OpenREM (OpenREM, 2025). This free software is used for the management and analysis of radiation doses in diagnostic imaging exams.

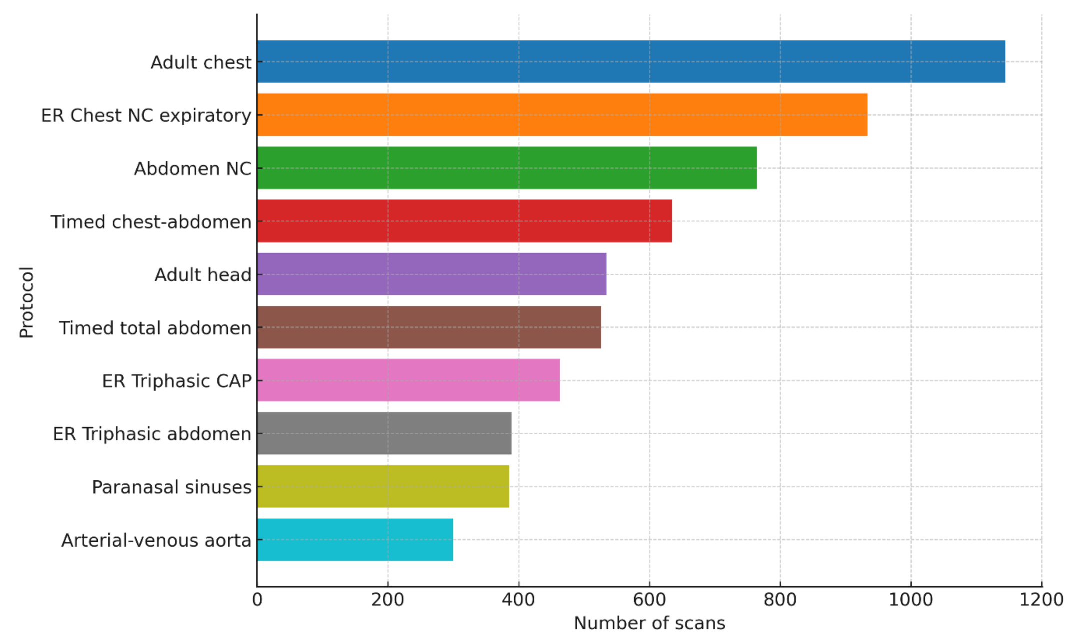
3. Results and Discussion

The analysis included over 3,700 CT exam records, representing a substantial volume of clinical data. Figure 1, generated using the OpenREM software, presents the protocols along with the corresponding number of exams and the respective average CTDIvol values.

Table 1. The 10 most frequently used CT protocols in the Radiology Department of HC-UFPE.

Protocol	Number of scans	$CTDI_{vol}$ (mGy)
Adult chest	1144	5.01
ER Chest NC expiratory	934	7.02
Abdomen NC	764	7.71
Timed chest-abdomen	634	9.20
Adult head	534	22.32
Timed total abdomen	526	11.70
ER Triphasic CAP	463	8.95
ER Triphasic abdomen	389	8.95
Paranasal sinuses	386	21.67
Arterial-venous aorta	300	12.30

Figure 1: The 10 most frequently used CT protocols in the CT sector of HC-UFPE.



It was observed that most protocols were applied to CT exams performed on patients with an average age above 50 years. Notably, the emergency triphasic chest-abdomen-pelvis (PS-TAP) and emergency triphasic abdomen (PS-Abdomen) protocols were more frequently used in exams conducted on male patients. Protocols such as paranasal sinuses and abdomen without contrast were more commonly applied to exams of patients with lower average ages, particularly among females, indicating these are more frequent in intermediate age groups.

Figure 2: Mean age by protocol and sex for the 10 most frequently used protocols.

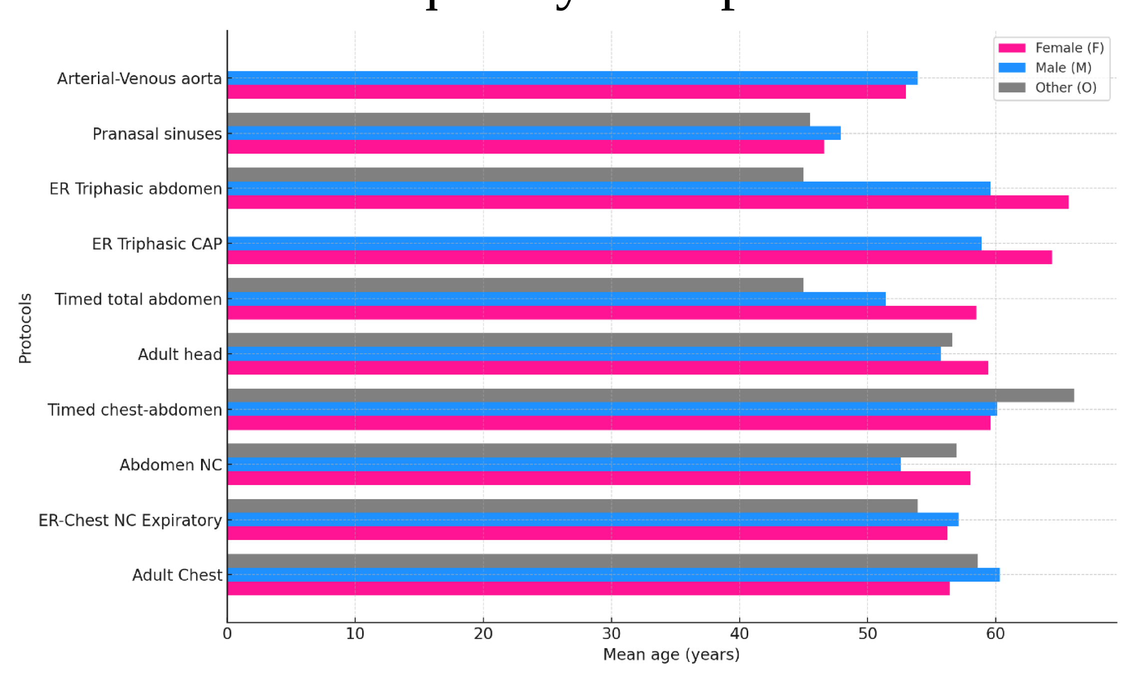


Table 2. BMI of the 10 most frequently used protocols in the HC-UFPE department.

Protocol	BMI Average (kg/m^2)
Adult chest	28.45
ER Chest NC expiratory	37.36
Abdomen NC	26.91
Timed chest-abdomen	27.71
Adult head	25.89
Timed total abdomen	26.95
ER Triphasic CAP	28.88
ER Triphasic abdomen	26.48
Paranasal sinuses	27.30
Arterial-venous aorta	Error

4. Conclusions

The use of the OpenREM software enabled a detailed understanding of the CT scan protocols employed within the department, highlighting more efficient pathways for initiating the optimization process. Therefore, if questions arise about prioritizing less conventional protocols during optimization, the medical physicist can base their decisions on objective data derived from the actual usage within the service. These actions are thus grounded in factual data and the analysis of real clinical practice.

References

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